

FRANKFURT HAHN, Germany

WMO: 106160

Lat: 49.9461N

Lon: 7.2644E

Elev: 503

StdP: 95.43

Time Zone: 1.00 (EUC)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.9	(c) -7.1	(d) -13.0	(e) 1.3	(f) -5.9	(g) -10.2	(h) 1.7	(i) -5.7	(j) 12.5	(k) 7.3	(l) 11.3	(m) 5.7	(n) 3.7	(o) 140	(p) 0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.5	28.2	18.5	26.1	17.7	24.2	16.9	19.5	25.8	18.6	24.4	17.8	22.7	3.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.5	13.3	21.2	16.8	12.7	20.4	15.9	12.0	19.7	57.6	25.8	54.8	24.4	52.1	22.7	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.0	8.8	7.8	-11.6	31.7	2.9	2.0	-13.7	33.1	-15.4	34.3	-17.0	35.4	-19.1	36.9
			-12.0	21.6	2.9	0.9	-14.1	22.3	-15.8	22.8	-17.5	23.3	-19.6	24.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	0.4	1.1	4.6	8.7	12.4	15.7	17.5	17.2	13.6	9.2	4.4	1.5
	DBStd	7.20	4.26	4.17	4.08	4.18	3.82	3.67	3.61	3.44	3.28	3.65	3.65	3.95
	HDD10.0	1309	297	248	171	73	19	2	0	0	5	59	170	264
	HDD18.3	3531	556	482	426	289	186	96	61	63	148	284	418	522
	CDD10.0	906	0	0	4	34	95	171	231	223	112	34	3	0
	CDD18.3	85	0	0	0	0	3	16	34	28	4	0	0	0
	CDH23.3	614	0	0	0	1	19	114	241	216	23	0	0	0
CDH26.7	152	0	0	0	0	2	24	58	67	1	0	0	0	
Wind	WSAvg	3.9	4.5	4.5	4.2	3.7	3.6	3.3	3.4	3.3	3.5	4.1	4.1	4.4
Precipitation	PrecAvg	711	62	55	51	43	60	64	67	65	54	59	62	69
	PrecMax	979	198	159	137	99	130	142	196	133	123	183	118	205
	PrecMin	539	8	6	5	0	14	17	27	15	14	10	2	5
	PrecStd	107	36	35	32	22	29	31	36	30	29	34	28	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	12.2	17.0	22.5	25.8	29.8	31.0	32.0	26.1	21.0	14.8	11.1
		MCWB	9.2	7.4	9.7	13.0	17.6	20.0	19.1	19.0	16.9	15.3	12.4	9.5
	2%	DB	9.2	9.7	14.5	20.1	22.9	26.5	28.1	28.2	23.2	17.9	12.3	9.8
		MCWB	8.1	6.9	8.6	12.0	15.6	18.1	18.4	18.5	16.4	14.2	10.6	8.7
	5%	DB	7.9	8.1	12.1	17.8	20.9	24.1	26.0	25.3	21.0	15.9	10.9	8.2
		MCWB	7.0	6.6	7.9	11.2	14.3	17.3	17.8	17.4	15.5	13.6	9.7	7.5
	10%	DB	6.1	6.9	10.2	15.5	19.0	22.1	24.0	23.2	19.0	14.1	9.2	6.9
		MCWB	5.4	5.7	7.3	10.3	13.5	16.3	16.8	16.7	14.8	12.4	8.3	6.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.8	8.7	10.7	14.4	18.7	20.8	20.7	20.4	18.5	16.1	12.9	10.1
		MCDB	10.2	10.4	14.2	20.3	23.8	26.8	27.7	27.5	23.2	19.0	14.1	10.6
	2%	WB	8.3	7.7	9.7	12.9	16.7	19.1	19.4	19.2	17.2	14.7	11.0	8.8
		MCDB	9.0	8.9	12.6	18.5	21.3	24.8	25.7	25.9	21.6	16.9	12.1	9.5
	5%	WB	7.0	6.7	8.8	11.7	15.4	17.9	18.5	18.3	16.2	13.8	9.8	7.5
		MCDB	7.8	8.0	11.4	16.4	19.5	22.8	24.4	23.9	20.0	15.7	10.9	8.2
	10%	WB	5.4	5.6	7.8	10.6	14.1	16.9	17.6	17.3	15.2	12.6	8.5	6.2
		MCDB	6.0	6.8	10.0	14.9	17.7	21.2	22.6	22.0	18.3	14.2	9.3	6.7
Mean Daily Temperature Range	5% DB	MDBR	4.0	4.8	6.5	8.1	8.2	8.7	8.5	8.4	7.4	5.9	4.2	3.9
		MCDBR	5.0	6.8	9.5	11.1	10.6	11.5	11.7	12.0	10.3	8.4	6.1	5.0
		MCWBR	4.6	5.0	5.6	5.7	5.6	5.9	5.1	5.5	5.4	5.5	4.8	4.4
	5% WB	MCDBR	4.9	5.7	7.9	10.0	9.6	10.2	10.2	10.1	8.9	7.7	5.6	4.7
		MCWBR	4.8	4.6	5.2	5.4	5.6	5.8	4.9	5.2	5.1	5.3	4.8	4.4
Clear-Sky Solar Irradiance	taub	0.301	0.327	0.357	0.383	0.390	0.392	0.391	0.387	0.361	0.348	0.322	0.299	
	taud	2.438	2.388	2.322	2.289	2.295	2.307	2.299	2.340	2.395	2.426	2.436	2.427	
	Ebn at Noon	749	812	845	860	868	866	860	846	832	775	710	693	
	Edh at Noon	62	82	104	120	125	124	123	113	96	79	62	54	
All-Sky Solar Radiation	RadAvg	0.79	1.50	2.67	4.21	4.96	5.51	5.22	4.47	3.35	1.93	0.89	0.61	
	RadStd	0.12	0.28	0.41	0.52	0.50	0.50	0.57	0.45	0.41	0.19	0.11	0.09	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	+0.38	N/A	N/A	N/A	N/A	N/A
Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page